

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

V Semester of B Sc Biotechnology Examination, November, 2019

Course code & Name: BT507: Entrepreneurship and IPR

Date: 09.11.2019 Day: Saturday Time: 01.30 p.m to 02.00 p.m Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.	20
1.	Which of the following statement is true for Biotechnology businesses? (i) it is R&D intensive (ii) requires high investment (iii) involves heavy risk (iv) all of the above	
2.	The government program which is aimed at entrepreneurship is: (i) Skill India (ii) Make in India (iii) Startup India (iv) Swacch Bharat Mission	
3.	The high-net-worth individual who invests in private companies are called as: (i) GVFL (ii) Angels (iii) IDBI (iv) SIDBI	
4.	Which among the following category of products requires highest investment for its production? (i) Luxury products/specialty products (ii) Semi luxury/shopping goods (iii) Convenience goods/perishable (iv) all of the above	
5.	“The respondents are asked questions and encouraged to speak or project their feelings, attitudes or motives of third person and not themselves” describes (i) survey technique (ii) depth interview (iv) focus groups (iv) projective technique	
6.	In Gujarat, many companies produce and market <i>Trichoderma</i> based biopesticides. Such a market is said to possess the characteristics of : (i) monopoly (ii) monopolistic competition (iii) oligopoly (iv) pure competition	

7.	The statement “Describes the rationale of how an organization creates, delivers and captures value” is applicable to: (i) business plan (ii) business model (iii) business concept (iv) business idea	
8.	GVFL is (i) a bank (ii) a venture finance entity (iii) an angel (iv) none of the above	
9.	RTI Act has potential to reduce _____ in India. (i) corruption (ii) pollution (iii) riots (iv) illiteracy	
10.	Following is NOT a characteristic of entrepreneurs: (i) Success and achievement (ii) Risk avoider (iii) Opportunity explorer (iv) Perseverance	
11.	“an asset that is pledged to the bank to cover any losses the bank might incur if the business is not able to repay what it owes to the bank” indicates (i) Collateral (ii) mortgage (iii) loan (iv) real asset	
12.	Which of the following is a secondary source of marketing research data: (i) experimental data (ii) observation (iii) salesman (iv) journals	
13.	In a product life cycle, sales for a product is highest during which of the following phases? (i) introduction (ii) growth (iii) decline (iv) maturity	
14.	The major product of Shanta biotechnics was (i) hepatitis B vaccine (ii) insulin (iii) penicillin (iv) <i>Azotobacter</i> biofertilizer	
15.	The liability arising out of mismanagement of a retirement plan or any other health and welfare plan offered by the company is called as: (i) Financial liability (ii) legal liability (iii) fiduciary liability (iv) personal liability	
16.	Competitor’s cost - (i) is a sign of price sensitivity (ii) is floor price (iii) specifies strength of the brand (iv) is cost incurred by the competitor for promotion of their products	
17.	Axio Biosolutions is a start-up involved in the production of: (i) probiotics (ii) wound dressing products (iii) biofertilizers (iv) plant tissue culture	
18.	‘Patan Patola’ is a double-sided saree from Gujarat. Such products originating from a particular territory of a country, a region or a locality in that territory is covered under which form of IPR? (i) patents (ii) geographical indications (iii) copy right (iv) industrial designs	
19.	Which among the following is NOT protected under the Indian Patent Act but covered under Copy Right Act? (i) Cell lines (ii) Gene (iii) Protein (iv) bioinformatics softwares	
20.	Gujarat government has set up biotechnology incubator at (i) Savli (ii) Vadodara (iii) Ahmedabad (iv) Surat	

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Date: 09.11.2019 Day: Saturday Time: 02.00 p.m to 04.30 p.m Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I		
Q - II Answer the following questions as directed		20
Q-II (A)	Answer any one of the following:	04
	(i) Discuss major characteristics of biotechnology as a business sector in India.	
	(ii) Who is an entrepreneur? Describe any three characteristics of an entrepreneur.	
Q-II (B)	Answer any two of the following:	06
	(i) Describe any one source of finance for biotechnology enterprises.	
	(ii) Write a note on: Business model	
	(iii) Discuss any two major problems faced by entrepreneurs in India.	
Q-II (C)	Answer any one of the following:	04
	(i) Enlist key sections of a business plan? Briefly describe the matter to be included under any one of the key sections of the business plan.	
	(ii) Describe the process for obtaining loan from banks for entrepreneurship.	
Q-II (D)	Describe any two of the following:	06
	(i) Giving advantages and disadvantages describe any one technique for market survey.	
	(ii) Give business idea for any biotechnology based product.	

SECTION - II		
Q-III	Answer the following questions as directed	30
QIII-(A)	Answer any one of the following: (i) Compare the characteristics of a private limited company and Partnership firm. (ii) Discuss Risk groups and Biosafety levels	04
Q-III (B)	Write notes on any two of the following:	06
	(i) Market segmentation (ii) Porter's five forces model (iii) Startup India	
Q-III (C)	Answer any one of the following:	04
	(i) What is risk? How risk in biotechnology enterprises can be mitigated? (ii) What is patent? Explain in one or two sentences the information provided by any two sections of a patent document.	
Q-III (D)	Answer any two of the following:	06
	(i) What is WTO? What are its objectives? (ii) Differentiate between copy right and industrial designs. (iii) Give value chain for any one of the product or service.	
Q-III (E)	Analyze the following biotech business case and answer the questions given below the case study:	10
	Pandorum Technologies, a Bengaluru-based biotech startup, has developed an artificial tissue that performs the functions of the human liver. Pandorum said these 3D printed living tissues made of human cells would enable affordable medical research with reduced dependence on animal and human trials. It will also eventually lead to full scale transplantable organs. Arun Chandru, 30-year-old co-founder of Pandorum, said liver toxicity and drug metabolism are the key hurdles, and contributors to failed human trials. Pandorum's 3D bio-printed mini-livers that mimic the human liver will serve as test platforms for discovery and development of drugs and vaccines. The firm said these drugs would have better efficacy, less side-effects and be developed at lower costs. "We developed everything here in India," said Mr. Chandru. "We can grow thousands of these tissues in the laboratory and test the efficacy of drugs on them for diseases including cancer." He said large pharma companies on an average spend about \$10 billion (Rs. 66,290 crore) and 10 years on research and development to get a single new drug to the market.	

Tuhin Bhowmick (age 34), another co-founder of Pandorum, said development of artificial organs has numerous clinical uses. The cell-based miniature organs can be used to develop bio-artificial liver support systems for preserving life in patients who have developed liver failure. “In the near future, such bio-printed organs will address the acute shortage of human organs available for surgical transplantation,” said Dr. Bhowmick, who holds a Ph.D. from the Indian Institute of Science.

Pandorum was founded by a group of friends in 2011 who were pursuing their higher studies at IISc. They came together to work on the development of artificial human organs after winning a business competition. Surviving initially on money from friends and family, the team approached the Department of Biotechnology with their vision. The company was awarded funding support by the Biotechnology Industry Research Assistance Council in 2012. The same year, the company got incubated by the Centre for Cellular and Molecular Platforms in Bengaluru. Mr. Chandru said they created the innovation with a funding of about Rs. 1 crore, more than half of which came from the government. The global artificial organ and bionics market is expected to reach \$38.75 billion (Rs 2.5 lakh crore) by 2020 at an estimated compound annual growth rate (CAGR) of 9.3% from 2014 to 2020, according to a study by Grand View Research. Pandorum’s ultimate aim is to make personalized human organs such as lungs, liver, kidney and pancreas on demand, according to Mr. Chandru. Pandorum’s innovation takes the area of making artificial organs to the next level. Bengaluru-based bioinformatics firm Strand Life Sciences founded by IISc. professors had earlier developed a virtual liver that mimics the functions of liver through software simulation. It is a predictive method that integrates data and insights for deeper understanding of the impact of a drug on the liver. The platform can predict the toxicity of several known drugs and toxins and explain the mechanism.

Questions: (correct answer to each question carries 02 marks)

1. What was the motivation for founders of Pandorum technologies?
2. What were the sources of finance for the Pandorum technologies?
3. How a formal education in biotechnology can help in bioentrepreneurship?
4. What are primary and secondary activities for Pandorum technologies in value chain model?
5. Suggest any product/service similar that offered by Pandorum technologies.

